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# Half-metallic properties for the $Ti_2YZ$ ($Y$ = Fe, Co, Ni, $Z$ = Al, Ga, In) Heusler alloys: A first-principles study

Xiao-Ping Wei, Jian-Bo Deng, Ge-Yong Mao, Shi-Bin Chu, Xian-Ru Hu*

*Department of Physics, Lanzhou University, Lanzhou 730000, People's Republic of China*

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ABSTRACT

Using the full-potential local orbital minimum-basis method, the $Ti_2$-based full-Heusler alloys are studied. The results show that these compounds exhibit a half-metallic behavior, however, in contrast to the conventional full-Heusler alloys, the full-Heusler alloys show a Slater–Pauling rule $M_t = Z_t - 18$ between the total spin magnetic moment ($M_t$) and valence electron concentration ($Z_t$) per unit cell. Low formation enthalpy implies these Heusler alloys can be fabricated experimentally. The origin of the gap in these half-metallic alloys are well understood. It is found that the half-metallic properties of $Ti_2$-based compounds are insensitive to the lattice distortion and a fully spin polarization can be obtained within a wide range of lattice parameters. This is favorable in practical application.



## 1. Introduction

In the last decade, the half-metallic materials, exhibiting a complete spin polarization at the Fermi level, have received growing widespread attention due to its realistic applications for spintronic devices [1]. Which offers opportunities for a new generation of devices combining standard microelectronics with spin-dependent effects such as nonvolatile magnetic random access memories and magnetic sensors [2]. The first predicted half-metallic material was the half-Heusler alloy NiMnSb found by de Groot and collaborators in 1983 [3]. Since then, much attention has been paid to the Heusler alloys for new candidates. Many Heusler alloys have been predicted theoretically to be half-metals, many experiments also were carried out to establish their magnetic and transport properties in quick succession [4–14].

Heusler alloys are represented in general by the generic formula $X_2YZ$, where $X$ and $Y$ are transition metal elements and $Z$ is a *s-p* element [15,16]. Usually, the Heusler alloys crystallize in a highly ordered $L2_1$ structure and the prototype of structure is the $Cu_2MnAl$ alloy, which belongs to the $Fm\overline{3}m$ space group. In past studies, most Cr, Mn, Ni and Co-based Heusler alloys have been theoretically predicted to be half-metals in the $L2_1$ structure [4,17–30]. Some alloys with $Li_2AgSb$-type structure were also investigated [31–33]. In particular, the structure containing two Ti atoms per f.u. has rarely been studied by the electronic structure calculations up to now. In present work, our research is mainly focused on the $Ti_2YZ$ ($Y$ = Fe, Co, Ni and $Z$ = Al, Ga, In) Heusler alloys with the $Li_2AgSb$-type structure [34].

In this paper, we present a study on a series of $Ti_2YZ$ ($Y$ = Fe, Co, Ni, $Z$ = Al, Ga, In) compounds with the same structure. In these compounds, up to now, no reports on the half-metallicity have been found but $Ti_2CoAl$ [35]. So, it is necessary to study systematically the electronic structure and magnetic properties of the $Ti_2YZ$ ($Y$ = Fe, Co, Ni, $Z$ = Al, Ga, In) alloys by density functional calculations. The present paper is organized as follows. In Section 3.1, the used crystal structure and the corresponding formation enthalpy are illustrated. In Section 3.2, we discuss the origin of the gap in these compounds. In Section 4, the Slater–Pauling (SP) behavior between the total moment and valence electron concentration is discussed. In Section 5, we present the bulk modulus and the effect of lattice constant. In Section 6, we analysis the role of *s-p* element for these Heusler alloys. Finally in Section 7, we summarize our results and conclusion.

## 2. Calculation details

We have carried out density functional calculations using the scalar relativistic version of the full-potential local-orbital (FPLO) minimum-basis band-structure method [36,37]. In this scheme, the scalar relativistic Dirac equation was solved. For the present calculations, the site-centered potentials and densities were expanded in spherical harmonic contributions up to $\ell_{max} = 12$. The Perdew–Burke–Ernzerhof 96 of the generalized gradient

* Corresponding author.
*E-mail address:* huxianru@lzu.edu.cn (X.-R. Hu).

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approximation (GGA) was used for exchange-correlation (XC) potential [38]. Accurate Brillouin zone integrations were performed using the standard special k point technique of the tetrahedron. We found that $20 \times 20 \times 20 = 8000$ k points were sufficient in all cases. For a self-consistent field iteration, the convergence was set to both the density ($10^{-6}$ in code specific units) and the total energy ($10^{-8}$ hartree).

## 3. Results and discussion

### 3.1. Crystal structure

In general, the cubic $X_2YZ$ Heusler alloys are found with $Cu_2MnAl$-type structure but also with the $Li_2AgSb$-type structure. For the studied alloys, we adopt the $Li_2AgSb$-type structure. The only difference is the single-atom site occupation, the nonequivalent 4a and 4c Wyckoff position is occupied by the two *X* atoms, and the *Y* and *Z* are located on 4b and 4d positions, respectively. This structure is similar to the *XYZ* alloys with $C1_b$ structure, and the difference is only the vacancy filled by an additional *X* atom in $T_d$ symmetry. It is reported if the nuclear charge of the *Y* element is larger than the one of the *X* element from the same period, this structure will be frequently observed. Furthermore, the structure may also appear in compounds comprised of transition metals from different periods [39]. In fact, these two structures may be hardly distinguishable by X-ray diffraction and much care has to be taken in the structural analysis, as they have the general fcc-like symmetry.

For the $Ti_2YZ$ ($Y$ = Fe, Co, Ni, $Z$ = Al, Ga, In) Heusler alloys, it might be of interest to pay close attention to those compounds because these $Ti_2$-based compounds were not reported in any experiment. Within the FPLO scheme, a structural optimization was performed to find the lattice parameter resulting in the minimum of the total energy. It was also confirmed that the ferromagnetic configuration is lower in energy than the non-spin polarized case for these compounds. An anti-parallel spin arrangement of the *sp* atom with respect to the *X* atoms in the cubic lattice is shown. Also we observed a similar conclusion as mentioned above [40]. Based on the conclusion, it is easy to estimate the superlattice structure of the Heusler compounds. In addition, the selectivity of occupation sites for our materials is also supported. Thus, in this paper $Li_2AgSb$-type structure with 4a, 4c, 4b and 4d Wyckoff positions occupied by two *X* atoms, the *Y* and *Z* atoms, respectively, is applied.

The formation enthalpy has to be considered before the results of electronic structure are analyzed, because the studied alloys still have not been prepared. The formation enthalpies of the $Ti_2YZ$ ($Y$ = Fe, Co, Ni, $Z$ = Al, Ga, In) alloys can be calculated from the following equation:

$$\Delta H(Ti_2YZ) = E(Ti_2YZ) - 2E(Ti) - E(Y) - E(Z)$$

where $E(Ti_2YZ)$, $E(Ti)$, $E(Y)$ and $E(Z)$ are the calculated equilibrium total energies of the $Ti_2YZ$ compound, Ti and Co with HCP structure, Al and Ni with FCC structure, Fe with BCC structure, Ga with A11 structure and In with TETR structure, respectively. The corresponding values are given as follow: −0.24 eV/atom, −0.24 eV/atom and 0.02 eV/atom for $Ti_2FeAl$, $Ti_2FeGa$ and $Ti_2FeIn$, −0.26 eV/atom, −0.27 eV/atom and −0.04 eV/atom for $Ti_2CoAl$, $Ti_2CoGa$ and $Ti_2CoIn$, −0.26 eV/atom, −0.26 eV/atom and −0.08 eV/atom for $Ti_2NiAl$, $Ti_2NiGa$ and $Ti_2NiIn$, respectively. The low enthalpy implies $Ti_2YZ$ alloys can be fabricated experimentally.

### 3.2. Origin of the gap

In this section, we will focus on the energy gap in minority-spin states. Such a gap is the most important feature of a half-metal. In these $Ti_2$-based alloys, the calculated results show a band gap for

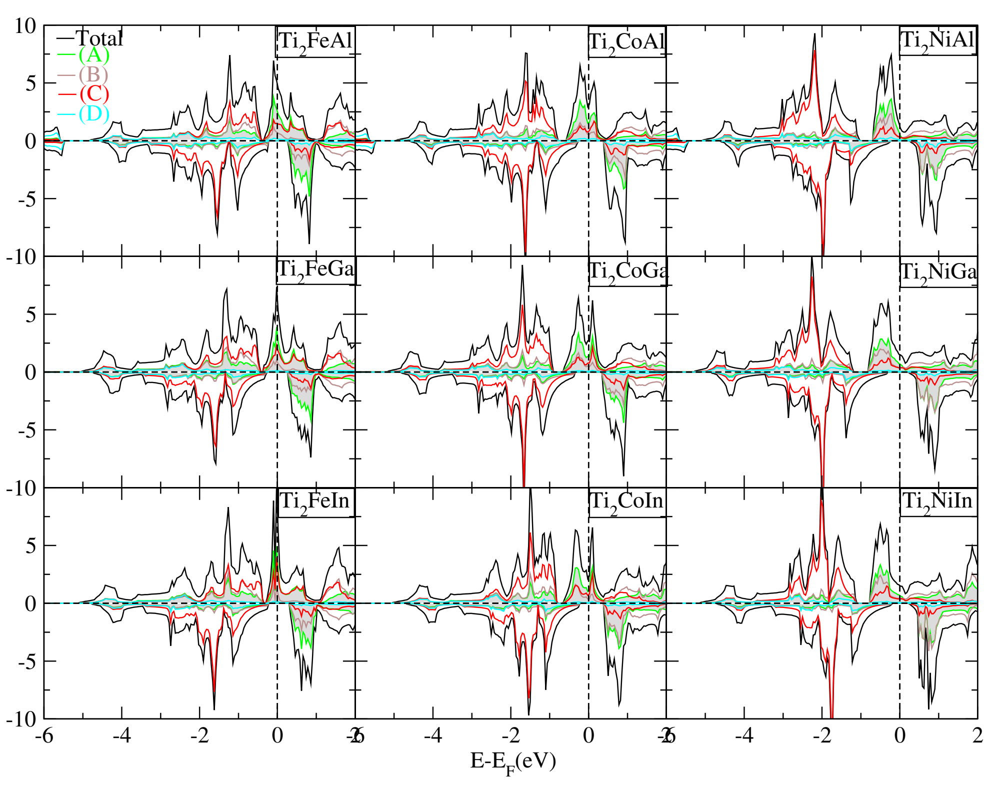


**Fig. 1.** Density of states for $Ti_2$-based alloys. Shown are the spin total and resolved DOS. (Note: upper and lower parts of the panels show for each compound the majority $\rho\uparrow$ and minority $\rho\downarrow$ DOS, respectively.)

the minority carriers, as shown in Fig. 1. The origin of the band gap is usually distinguished into three categories: (1) covalent band gap, (2) *d-d* band gap, and (3) charge transfer band gap [41]. The covalent band gap has been shown to exist in semi-Heusler alloys with $C1_b$ structure. The *d-d* band gap is responsible for the half-metallicity of the full-Heusler alloys with $L2_1$ structure. The charge transfer band gap is usually common in $CrO_2$ and double perovskites [33].

In our $Ti_2$-based alloys, the situation is rather complicated due to the special crystallized structure. The structure of $Ti_2$-based alloys has a space group similar to the semi-Heusler alloys but the empty site is occupied. Compared with the full-Heusler alloys with $L2_1$ structure, the Ti atoms occupy two sublattices with different surroundings. Therefore, it is important to pay attention to the origin of band gap.

To understand the origin of the band gap, let us observe the density of states (DOS) shown in Fig. 1. It can be seen that the $Ti_2$-based alloys with the same *Y* atom have the similar feature of DOS. The minority-spin DOS can be characterized by the large gap at the Fermi level and the occupied bonding states mainly present *Y* atom characters below the Fermi level. It is evident that the *d-d* orbitals hybridization between transition metals is rather intensive. Since the current studied Heusler alloys have a tetrahedral $T_d$ symmetry which is a subgroup of $O_h$, the possible hybridization between *d-d* orbitals sitting at the different sites will be created into several energy level, such as $e_g$, $t_{2g}$, $e_u$ and $t_{1u}$. Their complex exchange splitting will shift the Fermi level to the appropriate position and is responsible for the origin of gap. Furthermore, we can also see that the bonding hybrids are mainly localized at the high-valent transition metal atom such as Fe, Co or Ni, while the unoccupied anti-bonding states are mainly at the low-valent transition metal such as Ti. The bonding states preferentially reside at *Y*(*C*) sites and produce the spin density of occupied states. So, the covalent hybridization between high-valent and low-valent atoms also leads to the origin of gap [42].

As elucidated above, in these $Ti_2$-based alloys, the states near the Fermi level can be well ascribed to covalent hybridization and *d-d* orbitals hybridization between transition metals. In addition, as can be seen from the DOS patterns, the *Z* atom plays an important role in the half-metallicity of the Heusler alloys, although it does not directly form the minority gap, which provides *s* and *p* states to hybridize with *d* electrons and determines the degree of occupation of the *p-d* orbital. Thus, the hybridization between *p-d* electrons affects the formation of the energy gap and the width of gap. In conclusion, it should be emphasized that there exist two mechanisms in the formation of the band gap, that is covalent band gap and *d-d* band gap, but it is mainly the *d-d* band gap that characterizes the half-metallicity in $Ti_2$-based alloys.

## 4. Magnetic properties and Slater–Pauling rule

The preceding discussion is concerned with the electronic structure and the formation of the band gap. Now, we turn to the Slater–Pauling rule, which is a simple way to study the interrelation between the valence electron concentration and the magnetic moments (see Fig. 2). It is well known that the $C1_b$ Heusler alloys with three atoms per unit cell follow a simple rule that scales linearly with the number of valence electrons: $M_t = Z_t - 18$, where $M_t$ represents the total spin magnetic moment and $Z_t$ stands for the total number of valence electrons per unit cell, which was first noted by Kübler et al. [43] and the importance of which was also emphasized by Jung et al. [44] and Galanakis et al. [45] Since with 9 electron states occupied in the minority band for $C1_b$ Heusler alloys. $Z_t - 18$ is just the number of uncompensated electron spins. Namely, that is magnetic moment per unit cell.

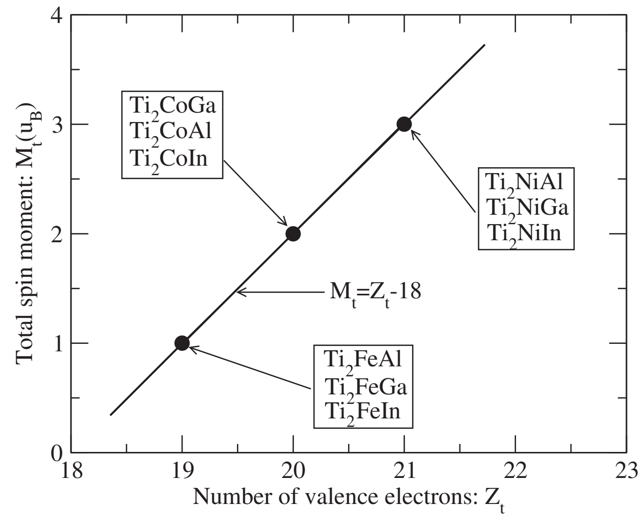


**Fig. 2.** Calculated total spin moments for the studied $Ti_2YZ$ ($Y$ = Fe, Co, Ni and $Z$ = Al, Ga, In) alloys. The solid line represents the Slater–Pauling curve for these Heusler alloys.

For ordered alloys with different kinds of atoms, it might be more convenient to work with all atoms of the unit cell. In the case of four atoms per unit cell, as in Heusler alloys like $Co_2MnGe$ with $L2_1$ structure, one has to subtract 24 from the accumulated number of valence electrons to find the magnetic moment per unit cell (*Mt*): $M_t = Z_t - 24$ [46] with $Z_t$ denoting the accumulated number of valence electrons in the unit cell containing four atoms. In the case of Heusler alloys, the number 24 arises from the number of completely occupied minority bands that has to be 12 in the half-metallic state. Particularly, these are one *s* ($a_{1g}$), three *p* ($t_{1u}$), and eight *d* bands. The latter consist of two triply degenerate bands with $t_{2g}$ symmetry and one with $e_g$ symmetry.

In Fig. 2 we have plotted a function between the number of valence electrons and spin magnetic moment per unit cell. The solid line represents the rule $M_t = Z_t - 18$ obeyed by these Heusler alloys. The total magnetic moment $M_t$ (in $\mu_B$) is just the difference between the number of occupied spin-up states and occupied spin-down states. The number of spin-down bands below the gap is in all cases $N_\downarrow = 9$. Thus, we can directly deduce the number $N_\uparrow = Z_t - N_\downarrow = Z_t - 9$ of occupied spin-up states and the moment, $M_t = (N_\uparrow - N_\downarrow)\ \mu_B = (Z_t - 2 \times N_\downarrow)\ \mu_B = (Z_t - 18)\ \mu_B$ from the total number of valence electrons ($Z_t$), is given by the difference. It might be with nine minority bands fully occupied. So, we can obtain the simple rule for these studied Heusler alloys: $M_t = Z_t - 18$. The calculated total moment $M_t$ is an integer 1, 2 and 3 $\mu_B$ with respect to total valence electrons 19, 20, 21. Which is in good agreement with our expected results. For all these studied Heusler alloys, we find that the total spin moment scales accurately with the total valence electrons and that they all present half-metallicity.

## 5. Bulk and effect of the lattice parameter

### 5.1. The changes of gap with lattice constant

In order to expound the influence of the lattice constant for width gap, we give the width of the minority gap with increasing the lattice parameter in Fig. 3. It can be seen that the increase of the lattice enlarges the energy gap for $Ti_2Fe$-based alloys, while the width gap of $Ti_2Ni$-based alloys is continuously decreasing with the expansion of lattice parameter. The change of the width of the gap can be traced back to two combined effects: (1) *p-d* orbital hybridization, which

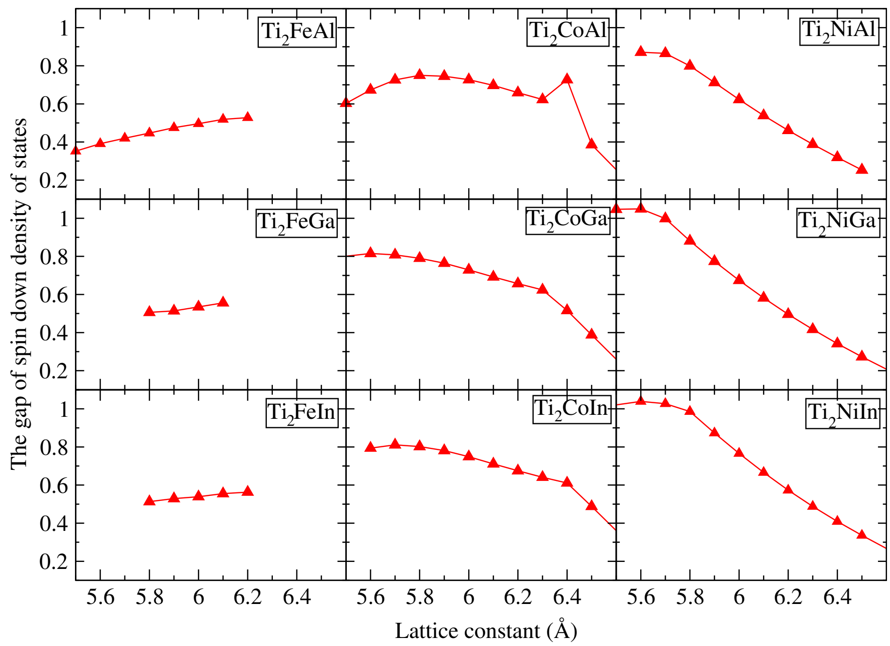


**Fig. 3.** The width of minority gap changes with the lattice parameters in the range of 5.50–6.60 Å.

obviously depends on the *p* states of *s*-*p* element and *d* states of *Y* atom. (2) their different lattice parameters. In Fig. 3, one can also see clearly the changes of width of the gap with different lattice parameters. From the partial DOS of atoms, it is clear that the binding energy of *p* states of *Z* atoms decreases generally with increasing atomic number, which seems well to illustrate the change of the gap as stated for $Ti_2Co$-based alloys. In fact, the effect of *p*-*d* hybridizations covers up the effect of lattice parameter, the contraction or expansion of the lattice parameter has an influence on the delocalized *p* electrons but not well localized *d* electrons of the transition metal. It is likely that the decreasing binding energy squeezes the energy gap. Furthermore, we also show bulk modulus and the range of pressure existing in the half-metallicity in Table 1. Changing the lattice parameter results in shifting the Fermi level without destroying the gap. The results show that the $Ti_2FeIn$ alloy is difficult to be compressed with the bulk modulus up to 264 GPa. The detailed values of the calculated total and atomic resolved moments are also included in Table 1.

## 6. The influence of the main element

In this Section, we will study on the role of the *s*-*p* element in the electronic properties of the studied Heusler alloys. Firstly, we focus on the influence of atomic number change of *s*-*p* element. Secondly, we consider further the influence of *s*-*p* element substituted by the empty site, where we performed calculations using the same lattice parameter as $X_2YZ$. In short, they are very important for the physical properties and the structural stability of the studied Heusler alloys, as we will discuss in the following. There are two important features:

(i) The influence of change of atomic number of *s*-*p* element. In Fig. 1, we can see clearly that the total DOS have almost similar feature with the change of *s*-*p* element, the hybridization between *p* states of *s*-*p* element and the *d* states of transition metal, which determines the degree of occupation of the *p*-*d* orbitals, has marginal differences. The energy of the *p* electrons is strongly dependent on *Z* element, which can be clearly seen in the resolved DOS patterns. The hybridization between *p* electrons and *d* electrons with different energies is crucial to the width of the energy gap in the minority-spin band. Even though the Fermi level is shifted slightly, which remains to be pinned in the gap of minority-spin states. Therefore, we predict that partly substituting one kind of *s*-*p* element for another might change the other properties of the material such as spin moment but not destroy their half-metallicity. In fact, the change of the *s*-*p* element results in a rigid shift of the bands [47].

(ii) The *s*-*p* element is very important for the structural stability, which can be seen obviously from the calculated DOS shown in Fig. 4, the DOS with *s*-*p* element shift slightly to bonding states improving the structural stability and is delocalized due to the strong exchange interaction, since metallic alloys prefer highly coordinated structures like fcc, bcc, hcp, etc. Therefore, the *s*-*p* elements are decisive for the structural stability of the studied Heusler alloys. A careful discussion of the bonding in the conventional Heusler alloys has been recently published by Nanda and Dasgupta [48] using the crystal orbital Hamiltonian population (COHP) method.

## 7. Summary and conclusions

In summary, we have studied in detail the electronic and magnetic properties of the $Ti_2$-based series Heusler alloys. It is shown that these Heusler alloys can be fabricated experimentally due to the low formation enthalpy and the gap is imposed by the lattice structure. The hybridization between covalent bonding in high-valence and lower-valence transition metal and *d*-*d* orbitals leads to the origin of the minority gap.

The calculated total magnetic moments $M_t$ are $1\mu_B$/f.u. for $Ti_2FeZ$ ($Z =$ Al, Ga, In), $2\mu_B$/f.u. for $Ti_2CoZ$ ($Z =$ Al, Ga, In), and $3\mu_B$/f.u. for $Ti_2NiZ$ ($Z =$ Al, Ga, In) following the $M_t = Z_t - 18$ rule, which agree with the Slater–Pauling curve quite well. The absence of an obvious variation in $M_t$ for different alloys is mainly ascribed to the counterbalance of the moments of *X* and *Y* in value and direction. The spin polarization of these alloys is found to be 100% within the pressure range, which is attractive in practical applications.

According to the influence of *s*-*p* element described, it is confirmed that the stability of the structure and the half-metallicity

**Table 1**
Magnetic data, bulk modulus and width of minority gap for series $Ti_2YZ$ ($Y$ = Fe, Co, Ni and $Z$ = Al, Ga, In) with optimized lattice parameter, and the range of pressure keeps the half-metallic properties is also listed.

| Compounds | $a_{opt}$ (Å) | $X_A$ ($\mu_B$) | $X_B$ ($\mu_B$) | $Y_C$ ($\mu_B$) | $Z_D$ ($\mu_B$) | $M_{tot}$ ($\mu_B$/f.u) | P (GPa) | $B$ (GPa) | Gap width (eV) |
|---|---|---|---|---|---|---|---|---|---|
| $Ti_2FeAl$ | 6.14 | 1.21 | 0.84 | −1.02 | −0.02 | 1.00 | 973.97 | 130.6 | 0.53 |
| $Ti_2FeGa$ | 6.12 | 1.22 | 0.93 | −1.09 | −0.06 | 1.00 | 401.50 | 125.6 | 0.56 |
| $Ti_2FeIn$ | 6.23 | 1.32 | 1.06 | −1.29 | −0.08 | 1.00 | 780.12 | 264.0 | 0.56 |
| $Ti_2CoAl$ | 6.14 | 1.50 | 0.80 | −0.23 | −0.07 | 2.00 | 899.55 | 137.4 | 0.68 |
| $Ti_2CoGa$ | 6.12 | 1.45 | 0.88 | −0.24 | −0.09 | 2.00 | 1197.55 | 78.8 | 0.68 |
| $Ti_2CoIn$ | 6.36 | 1.54 | 1.00 | −0.42 | −0.12 | 2.00 | 1209.13 | 129.3 | 0.62 |
| $Ti_2NiAl$ | 6.20 | 1.83 | 1.15 | 0.10 | −0.08 | 3.00 | 892.56 | 127.5 | 0.46 |
| $Ti_2NiGa$ | 6.19 | 1.78 | 1.24 | 0.10 | −0.11 | 3.00 | 962.82 | 125.3 | 0.50 |
| $Ti_2NiIn$ | 6.42 | 1.79 | 1.28 | 0.07 | −0.15 | 3.00 | 1208.90 | 119.6 | 0.39 |

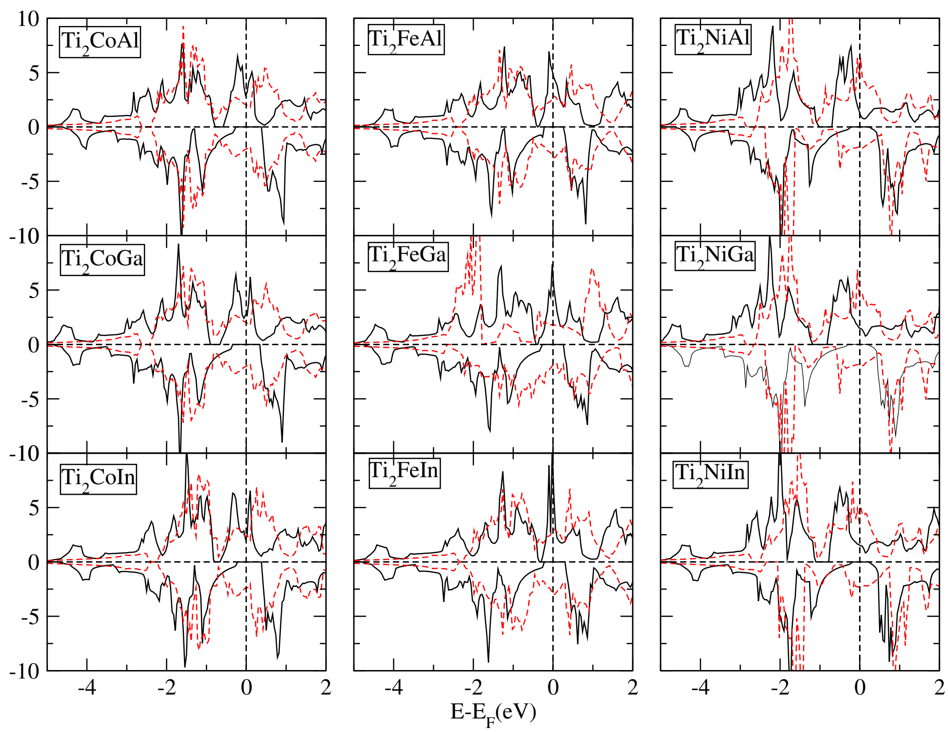


**Fig. 4.** The calculated total DOS for $Ti_2YZ$ and $Ti_2Y$ alloys. Of course we should keep in mind that all these calculations have been performed at the $Ti_2YZ$ lattice constants. In $Ti_2Y$ alloys, the $Z$ atom has been subsituted with a vacant site. Solid black line represents the total DOS of $Ti_2YZ$ alloys, and the total DOS of $Ti_2Y$ alloys are represented by red dotted line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of the $Ti_2YZ$ ($Y$ = Fe, Co, Ni and $Z$ = Al, Ga, In) Heusler alloys are dominated by the presence of *s*-*p* element. In addition, for all $Ti_2$-based Heusler alloys, the localized magnetic moment carried by the Ti and $Y$ atoms is restricted by the *s*-*p* element even though they have a small contribution to the total spin magnetic moment. Further, it is found that the minority gap of $Ti_2NiZ$ ($Z$ = Al, Ga, In) alloys shows a trend to decrease with increasing lattice parameter. The reason may be that the hybridization become weaker with the expansion of lattice parameter.

In conclusion, for these studied Heusler alloys, the special crystallized structure leads to a complex magnetic interaction and brings about many physical properties. Our findings suggest that these compounds can be viewed as a choice for further experimental investigations.

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